DropBox :

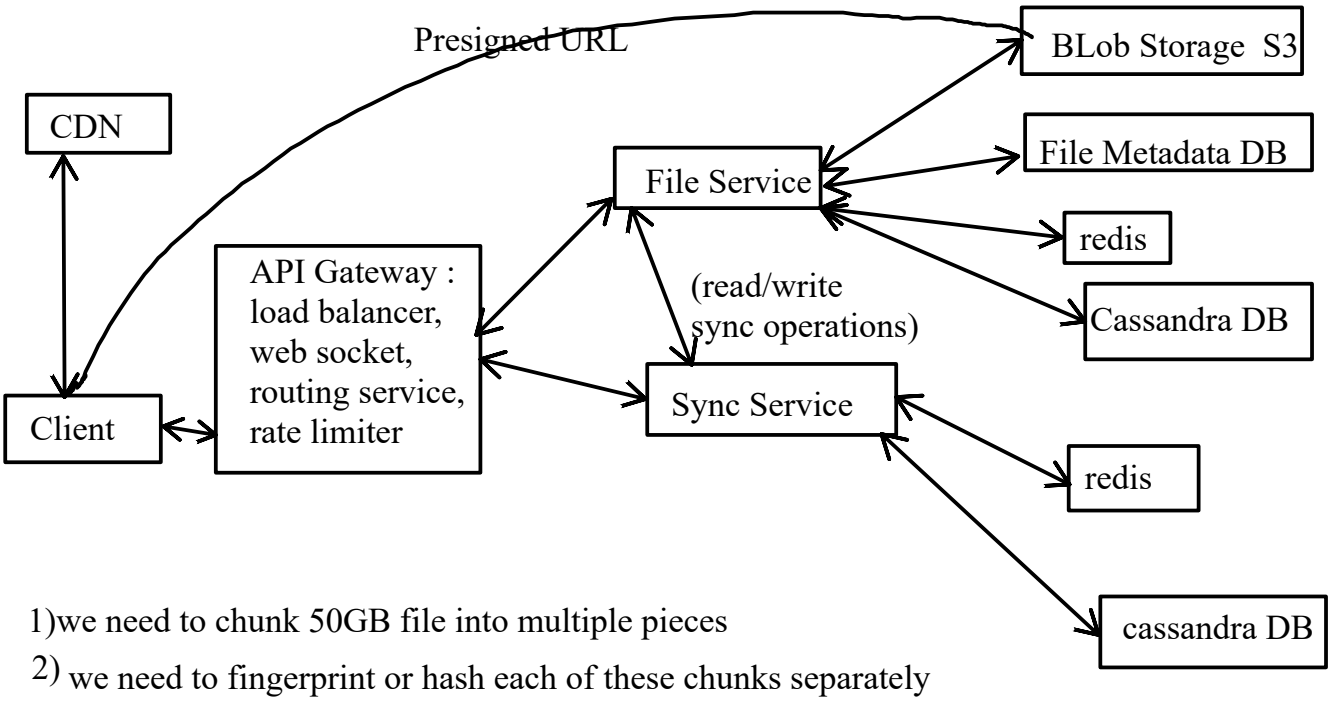
Functional requirements:
user should be able to upload/download any file based on his preference,
user should be able to search for any file based on his preference,

Non-functional requirements:
Data-partitioning,
load balancing,
fault tolerance,
vertical scalability,
horizontal scalability,
low latency,
high throughput,
CAP theorem (here eventual consistency is fine , but here we need 99.99% upitme i.e Availability>>Consistency)

Core Entities:
file ,
history,
filemetadata

API :
POST:/upaload/file
{
userID,
{filemetadata}
}
GET:/search/{fileID}

HLD :



1)we need to chunk 50GB file into multiple pieces
2) we need to fingerprint or hash each of these chunks separately
3)we need to upload each of these chunks separately
4) we need to update completed status for each of these chunks separately